42.5: Gas exchange occurs across specialized respiratory surfaces

In the remainder of this chapter, we will focus on the process of **gas exchange**. Although this process is often called respiratory exchange or respiration, it should not be confused with the energy transformations of cellular respiration. Gas exchange is the uptake of molecular O_2 from the environment and the discharge of CO_2 to the environment.

Partial Pressure Gradients in Gas Exchange

To understand the driving forces for gas exchange, we must consider **partial pressure**, which is simply the pressure exerted by a particular gas in a mixture of gases. Determining partial pressures enables us to predict the net movement of a gas at an exchange surface: A gas always undergoes net diffusion from a region of higher partial pressure to a region of lower partial pressure, down the pressure gradient. The rate of diffusion is proportional to the size of the gradient. In other words, an increase in the higher partial pressure or a decrease in the lower partial pressure will increase diffusion.

To calculate partial pressures, we need to know the pressure that a gas mixture exerts and the fraction of the mixture represented by a particular gas. Let's consider O_2 as an example. At sea level, the atmosphere exerts a downward force equal to that of a column of mercury (Hg) 760 mm high. Atmospheric pressure at sea level is thus 760 mm Hg. Since the atmosphere is 21% O_2 by volume, the partial pressure of O_2 is 0.21×760, or about 160 mm Hg. This value is called the *partial pressure* of O_2 (abbreviated PO_2) because it is the part of atmospheric pressure contributed by O_2. The partial pressure of CO_2 (abbreviated PCO_2) is much, much less, only 0.29 mm Hg at sea level.

Partial pressures also apply to gases dissolved in a liquid, such as water. When water is exposed to air, an equilibrium state is reached such that the partial pressure of each gas in the water equals the partial pressure of that gas in the air. Thus, water exposed to air at sea level has a PO_2 of 160 mm Hg, the same as in the atmosphere. However, the *concentrations* of O_2 in the air and water differ substantially because O_2 is much less soluble in water than in air **(Table 42.1)**. Furthermore, the warmer and saltier the water is, the less dissolved O_2 it can hold.

Table 42.1

Comparing Air and Water as Respiratory Media

A table titled Table 42.1 Comparing air and water as respiratory media.

Table 42.1 Comparing Air and water as Respiratory Media
O 2 partial pressure
O 2 concentration
Density
Viscosity

Respiratory Media

The conditions for gas exchange vary considerably, depending on whether the respiratory medium—the source of O_2—is air or water. As already noted, O_2 is plentiful in air, making up about 21% of Earth's atmosphere by volume. As shown in Table 42.1, air is much less dense and less viscous than water, so it is easier to move and to force through small passageways. As a result, breathing air is relatively easy and the exchange does not need to not be particularly efficient. Humans, for example, extract only about 25% of the O_2 in inhaled air.

Water is a much more demanding gas exchange medium than air. The amount of O_2 dissolved in a given volume of water varies but is always less than in an equivalent volume of air. Water in many marine and freshwater habitats contains only about 7 mL of dissolved O_2 per liter, a concentration roughly 30 times less than in air. Water's lower O_2 content, greater density, and greater viscosity mean that aquatic animals such as fishes and lobsters must expend considerable energy to carry out gas exchange. In the context of these challenges, adaptations have evolved that enable most aquatic animals to be very efficient in gas exchange. Many of these adaptations involve the organization of the surfaces dedicated to exchange.

Respiratory Surfaces

Specialization for gas exchange is apparent in the structure of the respiratory surface, the part of an animal's body where gas exchange occurs. Like all living cells, the cells that carry out gas exchange have a plasma membrane that must be in contact with an aqueous solution. Respiratory surfaces are therefore always moist.

The movement of O_2 and CO_2 across respiratory surfaces takes place by diffusion. The rate of net diffusion is proportional to the surface area across which it occurs and inversely proportional to the square of the distance through which molecules must move. In other words, gas exchange is fast when the area for

diffusion is large and the path for diffusion is short. As a result, respiratory surfaces tend to be large and thin.

In some relatively simple animals, such as sponges, cnidarians, and flatworms, every cell in the body is close enough to the external environment that gases can diffuse quickly between any cell and the environment. In many animals, however, the bulk of the body's cells lack immediate access to the environment. The respiratory surface in these animals is a thin, moist epithelium that constitutes a respiratory organ.

For earthworms, as well as some amphibians and other animals, the skin serves as a respiratory organ. A dense network of capillaries just below the skin facilitates the exchange of gases between the circulatory system and the environment. For most animals, however, the general body surface lacks sufficient area to exchange gases for the whole organism. The evolutionary solution to this limitation is a respiratory organ that is extensively folded or branched, thereby enlarging the available surface area for gas exchange. Gills, tracheae, and lungs are three such organs.

Gills in Aquatic Animals

Gills are outfoldings of the body surface that are suspended in the water. As illustrated in **Figure 42.21** (and in Figure 42.1), the distribution of gills over the body can vary considerably. Regardless of their distribution, gills often have a total surface area much greater than that of the rest of the body's exterior.

Figure 42.21

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    Diversity in the structure of gills, external body surfaces
that function in gas exchange.
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Diversity of the structure of gills and external body surfaces.

Part a. Marine worm. Body is highly segmented. Many polychaetes (marine worms of the phylum Annelida) have a pair of flattened appendages called parapodia on each body segment (here, parapodia appear as fringe on each side of each segment). The parapodia serve as gills and also function in crawling and swimming.

Part b. Crayfish. Crayfish and other crustaceans have long, feathery gills covered by exoskeleton (gills not visible in crayfish image). Specialized body appendages drive water over the gill surfaces beneath exoskeleton.

Part c. Sea star. The gills of a sea star are simple tubular projections of the skin. The hollow core of each gill is n extension of the coelom (body cavity). Gas exchange occurs by diffusion across the gill surfaces, and fluid

in the coelom circulates in and out of the gills, aiding gas transport. The tube feet surfaces also function in gas exchange.

Movement of the respiratory medium over the respiratory surface, a process called **ventilation**, maintains the partial pressure gradients of O_2 and CO_2 across the gill that are necessary for gas exchange. To promote ventilation, most gill-bearing animals either move their gills through the water or move water over their gills. For example, crayfish and lobsters have paddle-like appendages that drive a current of water over the gills, whereas mussels and clams move water with cilia. Octopuses and squids ventilate their gills by taking in and ejecting water, with the significant side benefit of getting about by jet propulsion. Fishes use the motion of swimming or coordinated movements of the mouth and gill covers to ventilate their gills. In both cases, a current of water enters the mouth of the fish, passes through slits in the pharynx, flows over the gills, and then exits the body **(Figure 42.22)**.

Figure 42.22

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The structure and function of fish gills.
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Blood in the capillaries of the gill filaments picks up oxygen by diffusion from water during countercurrent exchange, driven by the higher partial pressure of O_2 in the water.

Water flows into the fish mouth and over the gills (just behind the head; the operculum divides the head and body and protects the gills) before exiting the body. The gills are composed of repeating gill arches; each arch contains networks of blood vessels serving the two rows of gill filaments at each arch. A closer look at countercurrent exchange in the gill filaments reveals parallel blood vessels (the left conducting oxygen-rich blood and the right conducting oxygen-poor blood) with lamella (gill filaments) filled with capillaries between them. Water flows across the gill filaments perpendicular to the blood vessels and between sections of lamella. As water flows through the lamella, blood in the lamella capillaries picks up oxygen by diffusion from the water, driven by a partial pressure gradient for oxygen (partial pressure of O_2 is always higher in the water than in the blood).

A fish continuously pumps water through its mouth and over gill arches, using coordinated movements of the jaws and operculum (gill cover) for this ventilation. (A swimming fish can simply open its mouth and let water flow past its gills.) Each gill arch has two rows of gill filaments, composed of flattened plates called lamellae. Blood flowing through capillaries within the lamellae picks up O_2 from the water. Notice that the countercurrent flow of water and blood maintains a partial pressure gradient that drives the net diffusion of O_2 from the water into the blood over the entire length of a capillary.

Figure 42.22 Walkthrough

Structure and function of fish gills

Watch an explanation of this figure in the Figure 42.22 Walkthrough: The structure and function of fish gills.

image

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Animation

Gas Exchange in Fish Gills

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In fishes, the efficiency of gas exchange is maximized by **countercurrent exchange**, the exchange of a substance or heat between two fluids flowing in opposite directions. In a fish gill, the two fluids are blood and water. Because blood flows in the direction opposite to that of water passing over the gills, at each point in its travel blood is less saturated with O_2 than the water it meets (see Figure 42.22). As blood enters a gill capillary, it encounters water that is completing its passage through the gill. Depleted of much of its dissolved O_2, this water nevertheless has a higher PO_2 than the incoming blood, and O_2 transfer takes place. As the blood continues its passage, its PO_2 steadily increases, but so does that of the water it encounters, since each successive position in the blood's travel corresponds to an earlier position in the water's passage over the gills. The result is a partial pressure gradient that favors the diffusion of O_2 from water to blood along the entire length of the capillary.

Countercurrent exchange mechanisms are remarkably efficient. In the fish gill, more than 80% of the O_2 dissolved in the water is removed as the water passes over the respiratory surface. In other settings, countercurrent mechanisms contribute to temperature regulation and to the functioning of the mammalian kidney (see Concepts 40.3 and 44.4).

Tracheal Systems in Insects

In most terrestrial animals, respiratory surfaces are enclosed within the body, exposed to the atmosphere only through narrow tubes. Although the most familiar example of such an arrangement is the lung, the most common is the insect **tracheal system**, a network of air tubes that branch throughout the body. The largest tubes, called tracheae, open to the outside **(Figure 42.23)**. At the tips of the finest branches, a moist epithelial lining enables gas exchange by diffusion. Because the tracheal system brings air within a very short distance of virtually every body cell in an insect, the efficient exchange of O2 and CO2 does not require the participation of the animal's open circulatory system.

Figure 42.23

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A tracheal system.
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Grasshopper has repeating air sacks along the length of its body surrounded by network of branched internal tubes and trachea that open to the environment.

Part a. The respiratory system of an insect consists of branched internal tubes. The largest tubes, called tracheae, connect to external openings spaced along the insect's body surface. Air sacs formed from enlarged portions of the tracheae are found near organs that require a large supply of oxygen.

Part b. Air enters the trachea which branches into thinner tracheoles that attach to some of the body cells adjacent to an air sac. Rings of chitin keep the tracheae open, allowing air to enter and pass into smaller tubes called tracheoles. The branched tracheoles deliver air directly to cells throughout the body. Tracheoles have closed ends filled with fluid (blue-gray). When the animal is active and using more O2, most of the fluid is withdrawn into the body. This increases the surface area of air-filled tracheoles in contact with cells.

Part c. Mitochondria near multiple tracheoles. Muscle fibers also visible. Scale: 2.5 micrometers. The TEM above shows cross sections of tracheoles in a tiny piece of insect flight muscle. Each of the numerous mitochondria in the muscle cells lies within about 5 micrometers of a tracheole.

Tracheal systems often exhibit adaptations directly related to bioenergetics. Consider, for example, a flying insect, which consumes 10 to 200 times more O2 when in flight than it does at rest. In many flying insects, cycles of flight muscle contraction and relaxation pump air rapidly through the tracheal system. This pumping improves ventilation, bringing ample O2 to the densely packed mitochondria that support the high metabolic rate of flight muscle (see Figure 42.23).

Lungs

Unlike tracheal systems, which branch throughout the insect body, **lungs** are localized respiratory organs. Representing an infolding of the body surface, they are typically subdivided into numerous pockets. Because the respiratory surface of a lung is not in direct contact with all other parts of the body, the gap must be bridged by the circulatory system, which transports gases between the lungs and the rest of the body. Lungs have evolved both in organisms with open circulatory systems, such as spiders and land snails, and in vertebrates.

Among vertebrates that lack gills, the use of lungs for gas exchange varies. Amphibians rely heavily on diffusion across external body surfaces, such as the skin, to carry out gas exchange; lungs, if present, are relatively small. In contrast, most reptiles (including all birds) and all mammals depend entirely on lungs for gas exchange. Turtles are an exception; they supplement lung breathing with gas exchange across moist epithelial surfaces continuous with their mouth or anus. Lungs and air breathing also evolved in a few aquatic vertebrates as adaptations to living in oxygen-poor water or to spending part of their time exposed to air (for instance, when the water level of a pond recedes).

Mammalian Respiratory Systems: *A Closer Look*

In mammals, branching ducts convey air to the lungs, which are located in the *thoracic cavity*, enclosed by the ribs and diaphragm. Air enters through the nostrils and is then filtered by hairs, warmed, humidified, and sampled for odors as it flows through a maze of spaces in the nasal cavity. The nasal cavity leads to the pharynx, an intersection where the paths for air and food cross **(Figure 42.24)**. When food is swallowed, the **larynx** (the upper part of the respiratory tract) moves upward and tips the epiglottis over the glottis, which is the opening of the **trachea**, or windpipe. This allows food to go down the esophagus to the stomach (see Figure 41.9). The rest of the time, the glottis is open, enabling breathing.

Figure 42.24

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The mammalian respiratory system.
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Alveoli are at the bronchiole terminal ends. Oxygen-poor blood is delivered to the alveoli from the heart via the pulmonary artery; oxygen-rich blood returns via the pulmonary vein.

From the nasal cavity and pharynx, inhaled air passes through the larynx, trachea, and bronchi to the bronchioles, which end in microscopic alveoli lined by a thin, moist epithelium. Branches of the pulmonary arteries convey oxygen-poor blood to the alveoli; branches of the pulmonary veins transport oxygen-rich blood from the alveoli back to the heart.

The diaphragm is along the bottom of the lungs; the heart is positioned near the center of the chest, behind the lungs. Within the respiratory system, the trachea branches into the bronchi (left and right) which branch into the bronchioles. At the terminal ends of the bronchioles are the alveoli. Alveoli resemble clusters of grapes lined with a dense bed of capillaries (visible in a SEM at scale 50 micrometers); oxygen-rich blood returns to the heart from the alveoli via the pulmonary vein and oxygen-poor blood is delivered to the alveoli from the heart via the pulmonary artery.

Animation

The Human Respiratory System

image

Play

image

Play

From the larynx, air passes into the trachea. The cartilage that reinforces the walls of both the larynx and the trachea keeps this part of the airway open. Within the larynx of most mammals, the exhaled air rushes by a pair of elastic bands of muscle called vocal folds or, in humans, vocal cords. Sounds are produced when muscles in the larynx are tensed, stretching the cords so that they vibrate. High-pitched sounds result from tightly stretched cords vibrating rapidly; low-pitched sounds come from looser cords vibrating slowly.

The trachea branches into two **bronchi** (singular, *bronchus*), one leading to each lung. Within the lung, the bronchi branch repeatedly into finer and finer tubes called **bronchioles**. The entire system of air ducts has the appearance of an inverted tree, the trunk being the trachea. The epithelium lining the major branches of this respiratory tree is covered by cilia and a thin film of mucus. The mucus traps dust, pollen, and other particulate contaminants, and the beating cilia move the mucus upward to the pharynx, where it can be swallowed into the esophagus. This process, sometimes referred to as the “mucus escalator,” plays a crucial role in cleansing the respiratory system.

Gas exchange in mammals occurs in **alveoli** (singular, *alveolus*; see Figure 42.24), air sacs clustered at the tips of the tiniest bronchioles. Human lungs contain millions of alveoli, which together have a surface area of about 100 m2, 50 times that of the skin. Oxygen in the air entering the alveoli dissolves in the moist film lining their inner surfaces and rapidly diffuses across the epithelium into a web of capillaries that surrounds each alveolus. Net diffusion of carbon dioxide occurs in the opposite direction, from the capillaries across the epithelium of the alveolus and into the air space.

Lacking cilia or significant air currents to remove particles from their surface, alveoli are highly susceptible to contamination. White blood cells patrol the alveoli, engulfing foreign particles. However, if too much

particulate matter reaches the alveoli, the defenses can be overwhelmed, leading to inflammation and irreversible damage. For example, particulates from smoking or vaping enter the alveoli and can cause a permanent reduction in lung capacity. For coal miners, inhalation of large amounts of coal dust can lead to silicosis, a disabling, irreversible, and sometimes fatal lung disease.

The film of liquid that lines alveoli is subject to surface tension, an attractive force that has the effect of minimizing a liquid's surface area (see Concept 3.2). Given their tiny diameter (about 0.25 mm), alveoli would be expected to collapse under high surface tension. It turns out, however, that these air sacs produce a mixture of phospholipids and proteins called **surfactant**, for *surf*ace-*act*ive agent, which coats the alveoli and reduces surface tension.

In the 1950s, Mary Ellen Avery did the first experiment linking surfactant deficiency to *respiratory distress syndrome* (RDS), a life-threatening disease of newborns that at the time killed 10,000 infants annually in the United States **(Figure 42.25)**. RDS is most common among infants born before 32 weeks of gestation in the uterus. (The average full-term human pregnancy is 38 weeks.) Later studies revealed that surfactant typically appears in the fetal lungs after 33 weeks of gestation. Artificial surfactants are now used to treat early preterm infants, and treated babies with a body mass over 900 g (2 pounds) at birth usually survive without long-term health problems. For her contributions, Avery received the National Medal of Science.

Figure 42.25

Inquiry: What causes respiratory distress syndrome?

Experiment

Mary Ellen Avery, a research fellow at Harvard University, hypothesized that respiratory distress syndrome (RDS) in preterm infants is caused by the lack of a substance that reduces surface tension in the alveoli. To test this idea, she obtained autopsy samples of lungs from infants who had died of RDS or from other causes. She extracted material from each sample and let it form a film on water. Avery then measured the tension (in dynes per centimeter) across the water surface for each sample.

Results

Avery noted a pattern when she separated out samples from infants with body mass less than 1,200 g, the average mass of a fetus at 29–30 weeks of pregnancy.

Bar chart of surface tension of lung material by body mass. Infants with body mass over 1,200 grams who had died of RDS had the highest lung material surface tension.

Data from M. E. Avery and J. Mead, Surface properties in relation to atelectasis and hyaline membrane disease, *American Journal of Diseases of Children* 97:517–523 (1959).

Bar chart showing surface tension of lung material by body mass of infants who had died of RDS or other causes. The horizontal axis is body mass of infant; categories are less than 1,200 grams (n = 9 among infants who did not die from RDS and n= 0 among infants who died of RDS) and greater than 1,200 grams (n = 29 among infants who did not die from RDS and n = 9 among infants who died of RDS). The vertical axis is surface tension in dynes per centimeter, ranging from 0 to 40 with increments every 10. For infants with a body mass less than 1,200 grams, the sample included only those who had died from other (non RDS) causes. Surface tension in this group was about 25 dynes per centimeter. For infants with body mass greater than 1,200 grams, those who had not died of RDS had surface tensions much lower, or about 8 dynes per centimeter. Among those infants with body mass over 1,200 grams who had died of RDS, surface tension in the lung material was the highest, at just over 30 dynes per centimeter.

Conclusion

For infants with a body mass of 1,200 g or greater, samples from those who had died of RDS exhibited much higher surface tension than samples from those who had died from other causes. Avery inferred that infants' lungs normally contain a surface-tension-reducing substance (now called surfactant) and that RDS occurs when this substance is lacking. The results from infants with a body mass less than 1,200 g were similar to those from infants who had died from RDS, suggesting that surfactant is not normally produced until a fetus reaches this size.

WHAT IF? *If the researchers had measured the amount of surfactant in lung samples from the infants, what relationship would you expect between the amount of surfactant and infant body mass?*

*For suggested answer, see*Appendix A*.*

Having surveyed the route that air follows when we breathe, we'll turn next to the process of breathing itself.